



Review

The origins of the vaccine cold chain and a glimpse of the future[☆]John Lloyd^{*,1}, James Cheyne¹

7 Avenue Pasteur 71700 Tournus, France

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ABSTRACT

International efforts to eradicate smallpox in the 1960s and 1970s provided the foundation for efforts to expand immunization programmes, including work to develop immunization supply chains. The need to create a reliable system to keep vaccines cold during the lengthy journey from the manufacturer to the point of use, even in remote areas, was a crucial concern during the early days of the Expanded Programme on Immunization. The vaccine cold chain was deliberately separated from other medical distribution systems to assure timely access to and control of vaccines and injection materials. The story of the early development of the vaccine cold chain shows how a number of challenges were overcome with technological and human resource solutions. For example, the lack of methods to monitor exposure of vaccines to heat during transport and storage led to many innovations, including temperature-sensitive vaccine vial monitors and better methods to record and communicate temperatures in vaccine stores. The need for appropriate equipment to store and transport vaccines in tropical developing countries led to innovations in refrigeration equipment as well as the introduction and widespread adoption of novel high performance vaccine cold-boxes and carriers. New technologies also helped to make injection safer. Underlying this work on technologies and equipment was a major effort to develop the human resources required to manage and implement the immunization supply chain. This included creating foundational policies and a management infrastructure; providing training for managers, health workers, technicians, and others. The vaccine cold chain has contributed to one of the world's public health success stories and provides three priority lessons for future: the vaccine supply chain needs to be integrated with other public health supplies, re-designed for efficiency and effectiveness and work is needed in the longer term to eliminate the need for refrigeration in the supply chain.

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* Corresponding author.

E-mail addresses: john.lloyd1945@gmail.com (J. Lloyd), james@cheyne.net (J. Cheyne).

¹ Neither author is currently affiliated with an institution. Both are consultants.

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1. Introduction: Immunization in the 1960s and 1970s

Only a few vaccines were available in the early 1960s, and few children around the world received them. Smallpox was among the infectious diseases that were rampant, and the World Health Assembly received numerous reports of the catastrophic consequences of smallpox among its Member States. But vaccine technology existed for smallpox, offering the potential for protection.

In 1966, the World Health Organization (WHO) launched a global campaign to eradicate smallpox. This successful campaign demonstrated both the power and portability of vaccines. Within less than two decades, smallpox had been eradicated—a public health achievement that still stands as one of the greatest in history. Encouraged by the success of the smallpox campaign, health officials advocated for an expanded range of vaccines to be given routinely to infants under one year and women of child-bearing age.

In 1974, WHO established the Expanded Programme on Immunization (EPI), and Dr. Rafe Henderson became its first director shortly after. EPI was initially piloted in Ghana to assess the feasibility of establishing a single, global immunization schedule incorporating six antigens: tuberculosis, polio, diphtheria, pertussis, tetanus, and measles. The schedule was optimized to provide maximum protection for a minimum number of contacts through—what was then—a nascent primary health care system.

One of the key challenges of the early EPI work was to find a way to safely deliver vaccines, which are temperature-sensitive biological products, from the point of manufacture to the point of administration. Smallpox eradication established stepped vaccine distribution systems based on existing health services infrastructures but separate from the routine distribution of medicines. Recognizing the managerial weaknesses of medicine distribution at that time, WHO helped build the capacity of countries by developing the technologies, systems, and guidance towards a vaccine ‘cold chain’ to distribute vaccines routinely.

2. Challenges and solutions during development of the cold chain

In 1976, Professor David Morley of the Institute for Child Health, London, proposed that WHO establish a team within EPI to address three critical issues constraining WHO’s ambition to establish routine immunization services globally:

- An absence of systems to monitor the temperature of thermosensitive vaccines.
- An absence of appropriate equipment to store and transport vaccines.
- An insufficient number of adequately trained staff to handle vaccines.

WHO consultants prepared a strategy paper and a plan of action to tackle these issues by creating and disseminating appropriate technology and training materials for distribution and administration of vaccines [1]. The strategy envisaged separate ‘vaccine stores’ based on the typical pre-existing distribution hierarchy to ensure rapid implementation. Starting from the central or national store and ending at fixed, peripheral health facilities where immu-

nization services would be provided. This cold chain extended to periodic ‘outreach sessions’ held in communities that were far from the health facilities.

The vaccine distribution strategy also included injection and other supplies that are essential for the service. However, the strategy was targeted at immunization alone. Integration with medicines and other hospital supplies was rejected because the necessary control over stock management, transport priorities, maintenance and monitoring of storage temperatures could not be achieved at that time.

2.1. Absence of systems to monitor the temperature of thermosensitive vaccines

2.1.1. Challenges

All but one of the original EPI vaccines were sensitive to heat. Some were sensitive to freezing, although the extent of sensitivity was not fully known and freezing damage attracted little attention at the time. Because there was no way to assess the effects of heat exposure once the vaccines had been distributed, strict requirements ruled the process of vaccine handling and storage temperatures.

The standard procedure for temperature monitoring in 1976 was to read and record the temperature in each vaccine refrigerator twice daily and display the temperature profile on a chart each month. Large national stores had continuous temperature recorders that used a rotating disk of paper on which an ink stylus left a record of the temperature. Although compliance with standard procedures was good in some cases and action was taken when temperatures deviated beyond pre-set limits, compliance in other cases was poor and temperature reports were unreliable. The lack of systematic temperature monitoring also made it difficult to determine when cooling equipment required maintenance.

2.1.2. Solutions

From the beginning, WHO envisioned the need for an ‘end-to-end’ temperature monitoring system for vaccines in the cold chain. Beginning in the early 1980s, companies in the United States and Switzerland, including Berlinger & Co. AG, developed a cold chain monitor (CCM) based on blue wax absorption on a visual ‘track’. The CCM followed shipments of vaccine from manufacturer to countries and was used to monitor stores at all levels.

PATH (an international non-profit organization) and WHO began working in the late 1970s to find a way to track the heat exposure of individual vials of vaccine. Building on previous work to develop an enzyme indicator to warn of failures in the food cold chain in the United States, PATH and the Temptime Corporation developed and then commercialized a vaccine vial monitor (VVM) based on polymerization technology. VVMs are small stickers that adhere to vaccine vials and change colour irreversibly as the vaccine is exposed to heat, enabling health workers to easily determine whether the vaccine has been heat damaged [2]. WHO now requires that all vaccines purchased through the United Nations Children’s Fund (UNICEF) use VVMs.

The VVM solved a major problem presented by the absence of temperature monitoring, yet additional challenges remained. When and where did the temperature deviation occur? Was it the result of faulty equipment or poor practices? How would a

health worker know if a vaccine had been damaged by exposure to freezing?

Twenty years after the development of the CCM, the same Swiss company developed an electronic 30-day temperature recorder (30DTR). The 30DTR is now available as both a standalone recording device and a remote recording device providing alarms and data transmission with internet-based reporting.

For larger vaccine stores, electronic recorders have become the gold standard. Multi-channel temperature loggers with remote alarms and recording capabilities have made it possible to respond quickly to temperature alarms and use temperature data for analysis of refrigerator performance.

2.2. Absence of appropriate equipment to store and transport vaccines

2.2.1. Challenges

A formidable array of challenges existed in 1976 to ensure vaccines were kept at the recommended storage temperatures from arrival in the country to the point of use. Three challenges and their related solutions are presented as examples below.

Inadequate energy to power cooling equipment: Perhaps the most critical challenge was the scarcity and poor quality of energy available to refrigerate vaccines and to freeze ice packs for transport. In 1977, electricity was absent from approximately two-thirds of health facilities that stored vaccines, making it impossible to use standard electric-compression refrigerators. Only absorption-type refrigerators running on kerosene, gas, or electricity could be used. Even among facilities with access to an electrical grid, power supplies were intermittent and voltages fluctuated widely. Some countries used Diesel generators as an intermittent source, independent of the grid, but they required regular maintenance and were noisy and costly to run. Scarce, costly, and poor-quality grid electricity continues to plague many countries today.

Performance of refrigerators inadequate for vaccine storage and freezing icepacks: Absorption-type refrigerators were used widely in non-industrialized countries because they could be run on kerosene, gas, or electricity. But even with a clean fuel source and almost constant attention, the models available at that time could not maintain the temperature range required for vaccines or freeze enough ice packs for vaccine transport. More frequently, the fuel source in rural areas of developing countries was polluted with water, further reducing performance and increasing maintenance needs.

Short cold-life of passive-cooled containers: Insulated containers, including larger boxes for bulk vaccine and hand-carriers for immunization outreach activities, were needed to transport vaccine in high and low ambient temperatures. Picnic boxes and carriers cooled by frozen ice packs widely used in countries with temperate climates were unsuitable for transporting vaccine because their cold life was short. Their performance was not even sufficient for routine outreach lasting one or two days. They were also fragile and typically supplied with ice packs that were not interchangeable.

2.2.2. Solutions

To address the first and second challenges, the search for more reliable vaccine refrigeration equipment began with a preference for modifying low-cost refrigerators and freezers used to store food in the home. These 'Domestic' refrigerators and freezers were typically designed for industrialized markets where energy sources were reliable and the ambient temperature seldom exceeded +32 °C. To perform well enough to reliably store vaccines in areas of intermittent energy, polluted fuel supply and high and low ambient temperatures, modifications were made to the design of walls and cooling units. These performance-enhancing features were collected within draft standard performance specifications

from 1979. On this basis Electrolux developed a family of small refrigerators for health Centers that combined a standard absorption cooling unit with a new 25-l long-range cold box that had previously been developed for vaccine transport. More than three decades later, many of these refrigerators are still being supplied while the market for absorption refrigerators is disappearing.

Where grid electricity is provided, power cuts and 'brown-outs' can be frequent, even routine, in countries with problematic distribution. To maintain vaccine storage temperatures during power cuts, an engineering consultant from the Consumer Association Laboratory modified a domestic chest freezer, fitting it with an internal lining of water containers. After freezing the lining, the Ice-Lined Refrigerator (ILR) was set to maintain correct temperatures and remain stable with as little as eight hours of electricity per day. During the past decade, feedback from the field revealed a tendency for internal temperatures in ILRs to fall below freezing, and manufacturers have now modified the design to avoid freezing. Today the ILR is the vaccine storage device of choice because it guarantees continuous cooling wherever grid power is unreliable.

Refrigerators that are powered by solar (photovoltaic) energy are gradually replacing absorption refrigerators in the cold chain market. Several models of solar refrigerators were initially developed and deployed in the late 1980s, but problems with the battery and control module were common and difficult to fix. New-generation solar refrigerators, known as solar direct drive, adopted the concept of the ILR. Solar energy is used to freeze an ice liner that can keep the unit cold overnight or during cloudy days, so there is no need for a battery and control unit. The resulting refrigerator has demonstrated trouble-free operation without premature failure and now dominates sales.

To address the third and final challenge, helping immunization services reach further into rural areas, high-performance vaccine cold boxes and carriers were developed to transport vaccine. Starting with a wooden cold box developed by the National Bacteriological Laboratory in Sweden in 1974, Electrolux Luxembourg developed a transportable vaccine cold box that remained cool for more than five days at an ambient temperature of +43 °C. The Luxembourg cold box was tested successfully in Ghana and became a benchmark for many manufacturers in developing countries. Similarly, in 1977, a US manufacturer of portable containers collaborated with the Pan American Health Organization to reach performance objectives for outreach immunization. This vaccine carrier became the model for low-cost manufacturers in places like India and the Philippines to copy and improve. These carriers and boxes are now used throughout the developing world.

Three innovations enabled the countries to achieve successful results, driving the process of introduction and scale-up and ensuring the accuracy and credibility of manufacturers' performance claims:

- WHO technical staff and consultants collaborated directly on a one-to-one basis with many manufacturers and individual developers, particularly Electrolux, Temptime, and Berlinger. This intimate, mutual support made it easy to explain the special demands of extreme climates, weak infrastructure, low compliance, and the value of robustness in construction, and enabled manufacturers to achieve performance benchmarks.
- As a counterbalance to this public/private intimacy, engineering laboratories were used to test new products. These laboratories normally evaluated domestic and commercial products on behalf of retail consumers. The Consumers Association (UK) and Universidad del Valle (Colombia) acted as independent product design and testing contractors to immunization stakeholders.

- Through a procedure now known as Product Quality and Safety (PQS) WHO informed country management teams, UNICEF, and bilateral donors on equipment quality by publishing a list of pre-qualified, laboratory-evaluated equipment with guides and standard operating procedures.

Technologies listed in the WHO/PQS system were considered appropriate to meet a variety of programmatic needs and alleviated pressures on buyers to accept only the lowest price.

2.3. Insufficient number of adequately trained staff to handle vaccines

2.3.1. Challenges

Developing temperature monitoring and cooling equipment was a great technological challenge. Developing the management skills to operate and oversee the system was and remains an even greater challenge. The problem became evident following the eradication of smallpox, when countries began launching their national immunization programmes. At this stage, there was rarely a single manager in charge of immunization services. In addition, most programmes lacked mid-level managers or logisticians, and there were no consistent policies, procedures, or training materials that could be used as guidance.

2.3.2. Solutions

WHO pursued four solutions to human resource challenges related to the introduction of the EPI as a routine immunization service:

- The EPI was defined as a set of standard procedures with the common goal of achieving high coverage for the selected vaccines. In 1976, WHO distributed an EPI loose-leaf manual—known informally as the ‘Blue Book’—which became the basis for developing future policies and training.
- The supply and delivery infrastructure for vaccines was defined to meet the needs of the vaccines and the field operations of the EPI. Responsibility for reliable vaccine distribution was assigned to dedicated EPI managers. A cold chain simulation game was developed and used in several countries to demonstrate the complexities of vaccine distribution to senior managers in the Ministries and to help national EPI offices to develop and agree detailed logistics policies that best suited their needs.
- To address the global standard operating procedures of the EPI, staff at all levels of national immunization programmes received training and support. Starting in 1977, innovations such as participative training (replacing traditional presentations) were used to develop training courses for senior programme managers and regional- and district-level staff, and courses were developed in collaboration with the US Centers for Disease Control and Prevention. ‘Cascade training’ was offered to countries, typically starting with national-level training from where the participants were paired to manage training in their own regions and districts. Modules provided each participant with clear guidance with a focus on *what to do* after the end of the course. Special courses were developed for staff of other organizations that were active in immunization. These included courses held in Paris for Médecins Sans Frontières, in Atlanta for staff at the US Centers for Disease Control and Prevention, and, in the mid-1980s, for UNICEF staff based in regional and country offices.
- The final critical human resources solution was training targeted to the users and repair technicians of refrigeration equipment used to store vaccines. Because front-line health workers had little experience with compression refrigerators, they received additional support related to refrigerator care. Job aids were distributed, explaining what the health worker should do

daily, weekly, and monthly to care for and maintain their cold chain equipment. Refrigerator repair technicians received an intensive ten-day training course adapted from a nine-month course from a South London Technical College. Simple diagrammatic instructions were created, and after only ten days of hands-on practice, the participants could diagnose and repair all the typical failures of compression refrigerators. Course participants received a standard set of ‘universal spare parts’ developed by an engineer from Danfoss in Denmark that could be fitted to multiple models of refrigerators, along with a UNICEF-supplied toolkit, which is still available today.

In addition, WHO and other groups developed resources and software tools to help immunization programme staff with logistics planning and management. For example, vaccine stock control and forecasting was improved by applications such as the Commodities and Logistics Management (CLM) stock control tool created by John Snow, Inc. and Management Sciences for Health. CLM was extended to vaccines through a collaboration with WHO/EPI. By the mid-1990s, WHO had developed a more powerful, web-enabled vaccine stock control application known as the Vaccine and Supplies Management System, which is now used throughout the Eastern Mediterranean Region. More recently, several more stock control tools have been developed, and some are used widely.

3. Future directions for the immunization supply chain

The term ‘vaccine cold-chain’, adopted in 1976, has been replaced by ‘supply chain’ in the last five years. The name signals that the policy of exclusive distribution and storage of vaccine is evolving towards a strategy that encompasses both vaccines and medicines. Three factors explain this evolution towards a more integrated approach. First, logistics and distribution in health services have improved significantly in the last 25 years and both public and private systems are receiving more managerial attention. Second, the fastest growing demand is in the field of non-communicable diseases in many countries that require refrigerated storage either at +2 to +8°C or +20 to +25°C. Lastly, as the cost of procurement and operation of transport increases there is a more urgent need to rationalise supply trips, maximise utilization of vehicle capacity.

The design of the supply chain needs to be streamlined and improved as the number of vaccines continues to increase, the portfolio of immunization activities expands and new target groups are identified. Strategies of immunization are evolving towards greater integration. Routine immunization services given in fixed facilities, outreach immunization services carried to the field and periodic single-vaccine campaigns are merged into a common plan. Vaccine supplies and transport logistics should be managed together with other preventive services, particularly in remote immunization outreach context where the cost of reaching out beyond the ‘last-mile’ should be shared.

Other factors driving change in the supply chain include the growing use of information and communications technology, particularly mobile phones; better access to data related to vaccines and immunization; the need for much greater storage space to accommodate the growing number of vaccines; and the much higher costs for some newer vaccines, requiring that vaccine wastage rates be reduced.

Improvements in management, equipment, and transportation have enabled some countries to speed up vaccine distribution by reducing the number of steps in the system. While countries work to reduce the energy requirements of the supply chain, solar-powered refrigeration is already helping to ensure correct vaccine

storage in these areas. Solar roof-top arrays linked to the electrical grid have been shown to lower costs, increase security, and reduce environmental impact [3].

As supply chain performance improves, supported by the lessons learned over the past 40 years and particularly by the GAVI Alliance over the past 15 years, the delivery of vaccine should correspond more accurately to consumption, vaccine supply should become more regular, and the number of stock-outs should decline. Similarly, as efficiency of distribution receives more attention, risks associated with unpacking and repacking at each store should decline and less vaccine should expire before it reaches the intended recipient.

Managing changes to the supply chain and its continued development will require significantly more skilled managers to improve reliability and performance without any interruption of supplies. The International Association of Public Health Logisticians (IAPHL) and other organisations are promoting the professionalization of public health logistics through education, on-line forums and information sharing at a global level. This global awareness informs supply chain planning at country level, ensuring that country decisions are based on an evidence base and that changes are appropriate and 'outcome oriented'. While vaccine manufacturers and regulators work towards a future when the vaccine supply chain is without refrigeration, equipment developers and policy makers are now engaged in assessing and introducing improvements to achieve more reliable cooling and less burdensome procedures. The authors offer three priority goals for the next 25 years, as outlined below.

3.1. Extend, then eliminate the cold chain

Today, cold chain procedures and equipment continue to constrain the delivery of immunization services. Responding to this constraint, VVMs allow vaccines to be used to the limits of their stability when a cold chain failure occurs, but this implies off-label use. Vaccine is distributed within the cold chain as long and as far as cooling can be provided. Beyond the reach of the system and following breaks in the cold chain, vaccine can be used only if the VVM indication allows. Taking a vaccine deliberately out of the cold chain is only possible at present when that vaccine has been pre-qualified by WHO for 'Controlled Temperature Chain (CTC)'. The range of vaccines pre-qualified for CTC is growing but progress is slow and it seems today unlikely that all vaccines in the routine schedule will be able to taken out of the cold chain for many years. The Polio Eradication initiative (PEI) and several recent, successful trials of off-label vaccine use have shown increases in immunization coverage rates [4,5]. To reap this benefit the ultimate goal should be to eliminate refrigeration at +2°C to +8°C from the entire supply chain for all vaccines with corresponding on-label regulations for storage.

3.2. Provide faster, more reliable distribution

The current system of vaccine distribution is based on methods and dogma from 40 years ago. Vaccines are shipped through many levels (typically four up to six). Each level either collects fresh supplies of vaccines or receives vaccines that are delivered. This mix of collection and delivery leads to irregular, unreliable vaccine supply and poor-quality cold chain practices, including increased risk of vaccines freezing during transport.

High-performance management of the routine vaccine supply chain standardizes the delivery of vaccines to pre-set circuits of stores. The frequency of re-supply is set as high as possible, considering the risks of interruption of deliveries. The speed of distribution can also be increased by cutting the number of storage steps and optimizing the route chosen for each delivery circuit by choos-

ing optimal store locations and reducing 'reserve' stock levels. Pre-planned, reliable, and timely deliveries, combined with supervision of quantities supplied helps to eliminate stock-outs and accumulations of surplus stocks.

These programmatic benefits can be achieved by supply system redesign, not by the temporary, small-scale corrections that have characterised efforts to improve supply systems until recently.

3.3. Reach un-immunized populations with difficult access

The supply chain remains a constraint for reaching the final 15–20% of people who remain unimmunized. These people fall into one of three ranked priorities:

- The partially immunized who are unable to complete the series of contacts, often because of the quality of the service.
- Easily accessible populations, mostly living in peri-urban areas, who reject immunization or who cannot participate for various reasons.
- Those who live in rural areas that are physically difficult to access. The cost of service delivery per fully immunized child is much higher for this group than for the first two groups.

The first two groups are typically urban populations and are growing rapidly around the world. During the past decade, city populations have increased by about 750 million [6], and four-fifths of that growth has occurred in Africa and Asia [7]. By contrast, the third population group is rural and declining. These factors influence programmatic priorities and explain why people living in areas of difficult physical access are often reached last.

Remote populations—including those living on rivers, in mountains, on islands, and as nomadic tribes—often require a 'heroic' outreach system costing up to five times more than in urban areas to fully immunize a child [8]. In these remote areas, integrating immunization services with other selected services may help to share the cost of outreach. New technologies that streamline and extend the cold chain should also reduce costs and raise coverage.

4. Conclusion

As smallpox eradication was a stimulus for the EPI 40 years ago, the polio eradication initiative (PEI) is an important guide for transformative change in supply chains in the near future. The PEI has already successfully demonstrated:

- Using oral polio vaccine, the most heat-sensitive vaccine, off-label to the limits of its stability, guided by the VVM.
- Leapfrogging intermediate stores in many countries to speed up and streamline vaccine deliveries.
- Tracking children in the most remote areas where births are not registered. The PEI has established an enormous evidence base on reaching the unreached with the most heat sensitive vaccine in the schedule.

These and other initiatives that have contributed to the success of the PEI help to set the agenda for the development of immunization services, including all routine and supplementary delivery strategies for the short to medium term. The vaccine supply chain will first address the challenge of the 'last mile' by streamlining both the technologies to cool vaccines during the delivery of outreach services and the also to simplify standard operating procedures.

But as vaccine temperature sensitivity reduces and regulation permits, the opportunity exists to eliminate refrigeration in the vaccine supply chain. In the longer term the system will become

much less expensive and more easily managed. These procedures will help reach the unreached populations and enable faster distribution with less handling of vaccines.

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